A Physical Introduction to Higher Mathematics

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Contents

Part I

Waves

1 Linear Structure and Spectra

Definition 1.1 (Vector spaces and linear maps)**.** Let k be a field. A *k-vector space* is an abelian group V with an action $k \times V \to V$ satisfying distributivity, associativity, and $1v = v$. A map $T : V \to W$ is *linear* if $T(av + bw) = aT(v) + bT(w)$.

Definition 1.2 (Subspaces, kernels, images, and quotients)**.** A *subspace* of V is a subset closed under linear combinations. For a linear map $T : V \to W$, define

$$\ker T = \{v : T(v) = 0\}, \qquad \operatorname{im} T = \{T(v) : v \in V\}.$$

For a subspace $U \subseteq V$, the *quotient* V/U is the vector space of cosets $v + U$. The *direct sum* $V \oplus W$ is $V \times W$ with componentwise operations.

Exercise 1.1 (Universal linear constructions)**.** Prove that $\ker T$ and $\operatorname{im} T$ are subspaces. State and prove the universal properties of V/U and $V \oplus W$. Deduce $V/\ker T \cong \operatorname{im} T$ and the three isomorphism theorems.

Definition 1.3 (Duals and pairings)**.** The *dual* of V is $V^* = \operatorname{Hom}_k(V, k)$. A *pairing* of V and W is a bilinear map $V \times W \to k$. It is *nondegenerate* if each nonzero argument pairs nontrivially with some argument in the other space.

Definition 1.4 (Tensor products)**.** A *tensor product* of V and W is a vector space $V \otimes W$ with a bilinear map $(v, w) \mapsto v \otimes w$ such that every bilinear map $b : V \times W \to X$ factors uniquely through a linear map $\tilde{b} : V \otimes W \to X$. Tensor powers and multilinear maps are defined by iteration.

Exercise 1.2 (Duality and multilinearity)**.** For finite-dimensional spaces, construct the natural maps

$$V^* \otimes W \longrightarrow \operatorname{Hom}(V, W), \qquad V^* \otimes W^* \longrightarrow (V \otimes W)^*.$$

Prove that they are isomorphisms, identify the evaluation pairing, and prove the tensor–hom adjunction $\operatorname{Hom}(U \otimes V, W) \cong \operatorname{Hom}(U, \operatorname{Hom}(V, W))$.

Definition 1.5 (Inner products and orthogonality)**.** An *inner product* on a real vector space is a positive-definite symmetric bilinear form. On a complex vector space it is a positive-definite sesquilinear form, conjugate-linear in the first variable. Write $\|v\| = \sqrt{\langle v, v\rangle}$. Vectors are *orthogonal* when their inner product is zero, and $U^\perp = \{v : \langle u, v\rangle = 0 \text{ for every } u \in U\}$.

Definition 1.6 (Projections and adjoints)**.** A *projection* is an endomorphism P with $P^2 = P$. An *orthogonal projection* is a projection with $\ker P = (\operatorname{im} P)^\perp$. The *adjoint* of $T : V \to W$ is a map $T^* : W \to V$ satisfying $\langle Tv, w\rangle = \langle v, T^* w\rangle$.

Exercise 1.3 (Orthogonal decomposition and best approximation)**.** Let V be finite-dimensional and $U \subseteq V$. Prove $V = U \oplus U^\perp$, construct the orthogonal projection P_U, and prove that $P_U v$ is the unique element of U minimizing $\|v - u\|$. Prove that P is an orthogonal projection if and only if $P = P^* = P^2$.

Definition 1.7 (Isometries and spectral classes)**.** A linear map is *orthogonal* over $\mathbb{R}$, or *unitary* over $\mathbb{C}$, if it preserves the inner product. An endomorphism A is *self-adjoint* if $A = A^*$ and *normal* if $AA^* = A^*A$. A scalar λ is an *eigenvalue* if $\ker(A - \lambda\,\mathrm{id}) \neq 0$; a subspace U is *invariant* if $A(U) \subseteq U$.

Definition 1.8 (Spectral decomposition)**.** A *spectral decomposition* of an endomorphism A is an expression

$$A = \sum_{\lambda \in \mathrm{spec}(A)} \lambda P_\lambda, \qquad P_\lambda P_\mu = \delta_{\lambda\mu} P_\lambda, \qquad \sum_\lambda P_\lambda = \mathrm{id},$$

with orthogonal projections P_λ.

Exercise 1.4 (Finite-dimensional spectral theorem)**.** Prove that adjoints exist uniquely. Prove that a complex endomorphism is unitarily diagonalizable if and only if it is normal. Deduce spectral decompositions for self-adjoint, unitary, and normal operators, including the reality of self-adjoint spectra and the unit modulus of unitary spectra.

Exercise 1.5 (Diagonalization and invariant subspaces)**.** For an endomorphism with split characteristic polynomial, prove that it is diagonalizable if and only if the space is the direct sum of its eigenspaces. Show that normality removes generalized eigenvectors. Prove that commuting normal endomorphisms admit a common orthogonal spectral decomposition.

2 Oscillation and Fourier Analysis

Definition 2.1 (Linear oscillator)**.** A *linear oscillator* on a finite-dimensional real inner-product space V is an equation

$$M\ddot{q} + Kq = 0,$$

where M is positive definite and K is self-adjoint and positive semidefinite. A *normal mode* is a nonzero $v \in V$ for which $q(t) = a(t)v$ solves the equation for some nonconstant a.

Exercise 2.1 (The harmonic oscillator)**.** Solve $m\ddot{q} + kq = 0$ for $m, k > 0$. Prove conservation of $E = \frac{1}{2}m\dot{q}^2 + \frac{1}{2}kq^2$, determine the period, and deduce from the measured period of a small-amplitude mass–spring system the ratio k/m.

Exercise 2.2 (Coupled oscillators and normal modes)**.** Transform $M\ddot{q}+Kq = 0$ by $M^{1/2}$. Apply the spectral theorem to prove the existence of an M-orthogonal basis of normal modes and derive their frequencies. For a fixed-end chain of N equal masses joined by equal springs, compute the frequencies and mode shapes and identify the modes seen in a Chladni-type standing-wave measurement.

Definition 2.2 (Vibrating string)**.** For length L, tension τ, and linear density ρ, a *vibrating string* is a function $u : [0, L] \times \mathbb{R} \to \mathbb{R}$ satisfying fixed-end conditions $u(0, t) = u(L, t) = 0$ and

$$\rho\, \partial_t^2 u = \tau\, \partial_x^2 u.$$

Exercise 2.3 (String modes and measured pitch)**.** Derive the vibrating-string equation from transverse force balance in the small-slope limit. Determine all separated normal modes and their frequencies. Prove that the fundamental frequency is $f_1 = (2L)^{-1}\sqrt{\tau/\rho}$ and compare the predicted ratios $f_n/f_1 = n$ with the harmonics of a stretched string.

Definition 2.3 (Fourier function spaces)**.** For $1 \leq p < \infty$, the space $L^p(X, \mu)$ consists of measurable functions modulo equality almost everywhere with

$$\|f\|_p = \left(\int_X |f|^p \, d\mu\right)^{1/p} < \infty;$$

L^∞ uses the essential supremum. For Lebesgue measure on $\mathbb{R}^n$ and normalized arc measure on $\mathbb{T}$, write $L^p(\mathbb{R}^n)$ and $L^p(\mathbb{T})$. The inner product on $L^2(\mathbb{T})$ is $\langle f, g\rangle = (2\pi)^{-1}\int_0^{2\pi} \overline{f(x)}g(x)\,dx$. The exercises below establish the analytic properties used in this section.

Exercise 2.4 (Measure and integration)**.** Construct the integral from nonnegative simple functions. Prove monotone convergence, Fatou's lemma, dominated convergence, and Tonelli–Fubini.

Exercise 2.5 (L^p analysis)**.** Prove Hölder's and Minkowski's inequalities, completeness of L^p, and density of $C_c(\mathbb{R}^n)$ for $1 \le p < \infty$. Deduce that the inner-product norm makes L^2 complete.

Definition 2.4 (Orthogonal expansions)**.** Let H be an inner-product space. For an orthogonal family (e_n), the *orthogonal coefficients* of f are $\widehat{f}(n) = \langle e_n, f\rangle/\|e_n\|^2$. A *Fourier series* on $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ is the formal expansion $f \sim \sum_{n\in\mathbb{Z}} \widehat{f}(n)e^{inx}$, where $\widehat{f}(n) = (2\pi)^{-1}\int_0^{2\pi} f(x)e^{-inx}\,dx$.

Exercise 2.6 (Bessel, completeness, and Parseval)**.** Prove Bessel's inequality for every finite orthogonal family. Prove that an orthonormal family (e_n) is complete precisely when

$$\|f\|^2 = \sum_n |\langle e_n, f\rangle|^2$$

for every f. Prove completeness of $(e^{inx})_{n\in\mathbb{Z}}$ in $L^2(\mathbb{T})$ and obtain Parseval's identity and L^2 convergence of Fourier series.

Exercise 2.7 (Spectral evolution on an interval)**.** Expand suitable fixed-end initial data in string modes and derive the unique solution of the wave equation. Expand suitable initial temperature data in the same spatial eigenfunctions and derive the fixed-end solution of $\partial_t u = \kappa\partial_x^2 u$. Prove conservation of wave energy and monotone decay of heat energy directly from the spectral coefficients.

Definition 2.5 (Fourier transform and convolution)**.** For $f, g \in C_c(\mathbb{R}^n)$, define the *Fourier transform* and *convolution* by

$$\mathcal{F}f(\xi) = (2\pi)^{-n/2}\int_{\mathbb{R}^n} f(x)e^{-ix\cdot\xi}\,dx, \qquad (f * g)(x) = \int_{\mathbb{R}^n} f(y)g(x-y)\,dy.$$

The estimates $\|\mathcal{F}f\|_\infty \le (2\pi)^{-n/2}\|f\|_1$ and $\|f * g\|_1 \le \|f\|_1\|g\|_1$ extend these operations to the L^1 completion.

Exercise 2.8 (Fourier inversion, convolution, and Plancherel)**.** For Schwartz functions, prove Fourier inversion and $\mathcal{F}(f * g) = (2\pi)^{n/2}(\mathcal{F}f)(\mathcal{F}g)$. Prove that $\mathcal{F}$ preserves the L^2 inner product and extends uniquely to a unitary operator on $L^2(\mathbb{R}^n)$. Deduce the Plancherel theorem.

Exercise 2.9 (Wave and heat propagators)**.** Use the Fourier transform to solve the Cauchy problems for

$$\partial_t^2 u = c^2\Delta u, \qquad \partial_t u = \kappa\Delta u$$

on $\mathbb{R}^n$. Identify their spectral-mode evolution, derive the Gaussian heat kernel and its convolution law, and prove finite propagation speed for the one-dimensional wave equation.

Exercise 2.10 (Interference and Fourier optics)**.** For coherent scalar waves with complex amplitudes a_1, a_2, derive $|a_1 + a_2|^2$ and the fringe spacing in Young's double-slit geometry. In the Fraunhofer regime, derive that the far-field amplitude is the Fourier transform of the aperture. Compute the single-slit diffraction intensity, locate its zeros, and compare the predicted angular spacing with measured optical diffraction.

3 Hilbert Spaces and Spectral Operators

Definition 3.1 (Hilbert spaces and bounded operators)**.** A *Hilbert space* is a complete inner-product space. A linear map $T : H \to K$ is *bounded* if $\|T\| = \sup_{\|x\|\le 1} \|Tx\| < \infty$. Its adjoint is the unique bounded map satisfying $\langle Tx, y\rangle = \langle x, T^*y\rangle$. Projections, self-adjoint operators, and unitary operators satisfy respectively $P = P^* = P^2$, $A = A^*$, and $U^*U = UU^* = \mathrm{id}$.

Exercise 3.1 (Hilbert-space foundations)**.** Prove the projection theorem for closed subspaces and the Riesz representation theorem. Deduce $H = M \oplus M^\perp$ for closed M and prove existence, uniqueness, and $\|T^*\| = \|T\|$ for bounded adjoints.

Definition 3.2 (Projection-valued measures)**.** A *projection-valued measure* on a measurable space X assigns an orthogonal projection $E(B)$ to each measurable set, with $E(X) = \mathrm{id}$ and countable additivity in the strong operator topology. For bounded measurable f, the operator $\int_X f\, dE$ is defined by uniform approximation from simple functions.

Exercise 3.2 (Bounded spectral theorem)**.** Prove that every bounded self-adjoint operator A has a unique projection-valued measure on $\mathrm{spec}(A) \subset \mathbb{R}$ such that $A = \int \lambda\, dE(\lambda)$. Extend the statement to bounded normal operators, derive continuous functional calculus, and characterize unitary and projection spectra.

Definition 3.3 (Unbounded operators and domains)**.** A *densely defined operator* T has a dense linear domain $\mathrm{Dom}(T) \subset H$. It is *closed* when its graph is closed in $H \oplus H$; its graph norm is $\|x\|_T^2 = \|x\|^2 + \|Tx\|^2$. The adjoint T^* is defined on those y for which $x \mapsto \langle Tx, y\rangle$ is bounded on $\mathrm{Dom}(T)$. The operator is *symmetric* when $\langle Tx, y\rangle = \langle x, Ty\rangle$ on its domain and *self-adjoint* when $T = T^*$, including equality of domains.

Exercise 3.3 (Sobolev domains on an interval)**.** Show that $H^1([0, L]) = \{f \in L^2 : f' \in L^2\}$, with f' interpreted weakly, is complete. Prove that every class has an absolutely continuous representative, endpoint evaluation is continuous, and integration by parts holds. Identify H_0^1 as the closure of $C_c^\infty(0, L)$.

Exercise 3.4 (Self-adjoint momentum and boundary holonomy)**.** On $L^2([0, L])$, let $P_\theta = -i\hbar\, d/dx$ on the absolutely continuous functions with derivative in L^2 and boundary condition $\psi(L) = e^{i\theta}\psi(0)$. Use integration by parts and the adjoint domain to prove that P_θ is self-adjoint. Find its orthonormal eigenbasis and show

$$p_n = \frac{\hbar}{L}(2\pi n + \theta), \qquad n \in \mathbb{Z}.$$

Interpret θ as a boundary holonomy, later realized by magnetic flux through a ring, and explain why changing the domain changes the measured spectrum.

Exercise 3.5 (Unbounded spectral theorem)**.** Use the Cayley transform and the bounded spectral theorem to prove that every self-adjoint H has a unique projection-valued measure with

$$H = \int_{\mathbb{R}} \lambda\, dE_H(\lambda), \qquad \mathrm{Dom}(H) = \left\{\psi : \int_{\mathbb{R}} \lambda^2\, d\langle \psi, E_H(\lambda)\psi\rangle < \infty\right\}.$$

Construct the measurable functional calculus on its natural domains.

Exercise 3.6 (Stone's theorem)**.** Prove that $U_t = e^{-itH/\hbar}$ is a strongly continuous unitary group for every self-adjoint H. Conversely, recover a unique self-adjoint generator H from every strongly continuous unitary group and identify its domain by the strong derivative at $t = 0$.

Exercise 3.7 (Fourier spectra and free quantum evolution)**.** Use the Fourier transform to realize $-i\hbar\partial_j$ and $-\Delta$ as multiplication by $\hbar\xi_j$ and $|\xi|^2$ on their natural domains. Deduce self-adjointness, identify their spectral projections, and derive the free Schrödinger evolution for $H = -(\hbar^2/2m)\Delta$. Prove preservation of the domain, norm, and expected energy, and use this concrete multiplication-operator model to verify the preceding spectral and Stone theorems, including their domain statements.

Exercise 3.8 (Spectral lines from unitary evolution)**.** If $H|n\rangle = E_n|n\rangle$, use $U_t = e^{-itH/\hbar}$ to show that $\langle A(t)\rangle$ contains frequencies $(E_m - E_n)/\hbar$. Explain why spectroscopic peaks measure energy differences and why line strengths depend on matrix elements between the corresponding spectral subspaces. Compare with the atomic spectrum exercise below.

Part II

Symmetry

4 Groups and Representations

Definition 4.1 (Groups, homomorphisms, and quotients)**.** A *group* is a set G with an associative multiplication, an identity, and inverses. A *homomorphism* $\phi : G \to H$ preserves multiplication. A subgroup $N \leq G$ is *normal* if $gNg^{-1} = N$ for every $g \in G$; the *quotient group* G/N consists of cosets with induced multiplication.

Exercise 4.1 (The group isomorphism theorems)**.** Prove that kernels are normal and that a normal subgroup is the kernel of the quotient homomorphism. Prove the three isomorphism theorems for groups and the universal property of G/N.

Definition 4.2 (Actions, orbits, and stabilizers)**.** A *left action* of G on X is a homomorphism $G \to \mathrm{Bij}(X)$. The *orbit* and *stabilizer* of $x \in X$ are $Gx = \{gx : g \in G\}$ and $G_x = \{g : gx = x\}$.

Exercise 4.2 (Orbit–stabilizer and class equation)**.** Construct a natural bijection $G/G_x \to Gx$. For finite G, deduce the orbit–stabilizer formula and the class equation from the conjugation action. Use these results to prove that every group of prime order is cyclic.

Definition 4.3 (Representations and intertwiners)**.** A *representation* of G on V is a homomorphism $\rho : G \to \mathrm{GL}(V)$. An *intertwiner* $T : (V, \rho) \to (W, \sigma)$ satisfies $T\rho(g) = \sigma(g)T$. A subspace is *invariant* if it is preserved by every $\rho(g)$; a nonzero representation is *irreducible* if it has no proper nonzero invariant subspace. Direct sums and tensor products carry the actions $\rho \oplus \sigma$ and $g \mapsto \rho(g) \otimes \sigma(g)$.

Exercise 4.3 (Maschke's theorem)**.** Let G be finite and let the characteristic of k not divide $|G|$. Average an arbitrary projection over G to construct an invariant complement to every invariant subspace. Deduce that every finite-dimensional representation is a direct sum of irreducible representations.

Exercise 4.4 (Schur's lemma)**.** Prove that a nonzero intertwiner between irreducible complex finite-dimensional representations is an isomorphism and that every endomorphism of an irreducible representation is scalar. Deduce uniqueness of irreducible multiplicities in a decomposition.

Definition 4.4 (Characters)**.** The *character* of a finite-dimensional representation V is the class function $\chi_V(g) = \mathrm{Tr}(\rho_V(g))$. For class functions on a finite group, define

$$\langle f, h\rangle_G = \frac{1}{|G|}\sum_{g\in G} \overline{f(g)}h(g).$$

Exercise 4.5 (Character orthogonality and decomposition)**.** Use averaging of linear maps and Schur's lemma to prove orthonormality of the irreducible characters. Prove that they form an orthonormal basis of the class functions and that $\langle \chi_U, \chi_V\rangle_G = \dim \mathrm{Hom}_G(U, V)$. Deduce the multiplicity formula and the character table constraints.

Exercise 4.6 (Symmetry projections)**.** For an irreducible character χ of a finite group, prove that

$$P_\chi = \frac{\dim \chi}{|G|}\sum_{g\in G} \overline{\chi(g)}\rho(g)$$

is the projection onto the χ-isotypic subspace. Apply these projections to the displacement space of a molecule whose equilibrium configuration is preserved by G.

5 Continuous Symmetry and Angular Momentum

Definition 5.1 (Lie groups and Lie algebras)**.** A *Lie group* is a smooth manifold G whose multiplication and inversion are smooth. Its *Lie algebra* is $\mathfrak{g} = T_eG$ with bracket induced by left-invariant vector fields. A *one-parameter subgroup* is a smooth homomorphism $\mathbb{R} \to G$. The *exponential map* $\exp : \mathfrak{g} \to G$ sends X to the value at 1 of the one-parameter subgroup with initial velocity X.

Exercise 5.1 (One-parameter subgroups and the exponential map)**.** Prove existence and uniqueness of the one-parameter subgroup generated by $X \in \mathfrak{g}$. For a closed matrix group, prove that it is $t \mapsto e^{tX}$, identify the commutator bracket, and compute the first nontrivial terms of the Baker–Campbell–Hausdorff formula.

Definition 5.2 (Infinitesimal representations)**.** For a smooth representation $\rho : G \to \mathrm{GL}(V)$, its *infinitesimal representation* is

$$d\rho : \mathfrak{g} \to \mathrm{End}(V), \qquad d\rho(X) = \left.\frac{d}{dt}\right|_0 \rho(\exp tX).$$

Exercise 5.2 (Differentiating symmetry)**.** Prove that $d\rho$ preserves brackets and that $\rho(\exp X) = \exp(d\rho(X))$. Prove that a connected Lie-group representation is determined by its infinitesimal representation whenever the latter integrates uniquely.

Definition 5.3 (Rotations and spin)**.** Define

$$\begin{aligned} SO(3) &= \{R \in \mathrm{End}(\mathbb{R}^3) : R^*R = \mathrm{id},\ \det R = 1\},\\ SU(2) &= \{U \in \mathrm{End}(\mathbb{C}^2) : U^*U = \mathrm{id},\ \det U = 1\}. \end{aligned}$$

An *angular-momentum representation* is a unitary representation of $SU(2)$; with Planck constant $\hbar$, its infinitesimal generators J_1, J_2, J_3 are normalized by $[J_i, J_j] = i\hbar\varepsilon_{ijk}J_k$.

Exercise 5.3 ($SU(2)$ double-covers $SO(3)$)**.** Let $SU(2)$ act by conjugation on the real space of traceless Hermitian 2×2 operators. Prove that the resulting homomorphism $SU(2) \to SO(3)$ is surjective with kernel $\{\pm\,\mathrm{id}\}$. Deduce the effect of a 2π rotation in integer- and half-integer-spin representations.

Exercise 5.4 (Irreducible representations of $SU(2)$)**.** Using raising and lowering operators, prove that every finite-dimensional irreducible complex representation is indexed by $j \in \{0, \frac{1}{2}, 1, \ldots\}$, has dimension $2j+1$, and has J_3-eigenvalues $m\hbar$ for $m = -j, -j+1, \ldots, j$. Compute the spectrum of $J^2 = J_1^2 + J_2^2 + J_3^2$.

Exercise 5.5 (Clebsch–Gordan decomposition)**.** Prove

$$V_{j_1} \otimes V_{j_2} \cong \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} V_j.$$

Construct coupled angular-momentum eigenstates from highest-weight vectors and compute the singlet and triplet decomposition of $V_{1/2} \otimes V_{1/2}$.

Definition 5.4 (Quantum symmetries)**.** For a Hilbert space H and self-adjoint Hamiltonian $\mathcal{H}$, a *unitary symmetry* is a unitary U with $U\,\mathrm{Dom}(\mathcal{H}) = \mathrm{Dom}(\mathcal{H})$ and $\mathcal{H}U\psi = U\mathcal{H}\psi$ on that domain. A *quantum number* is an eigenvalue of a self-adjoint operator commuting with $\mathcal{H}$ in this sense. An operator A has *symmetry type* W when its components define an intertwiner from W to $\mathrm{End}(H)$ under conjugation.

Exercise 5.6 (Symmetry, degeneracy, and selection rules)**.** Prove that every eigenspace of $\mathcal{H}$ is symmetry-invariant and decomposes into irreducibles. Deduce symmetry-forced degeneracies. Prove that a transition matrix element $\langle f, Ai \rangle$ can be nonzero only if the trivial representation occurs in $V_f^* \otimes W \otimes V_i$, and derive the angular-momentum rules for an electric-dipole operator.

Exercise 5.7 (Atomic spectra)**.** For the Coulomb Hamiltonian, use rotational symmetry and separation into irreducible $SO(3)$-types to obtain the quantum numbers ℓ, m and their degeneracies. With the radial spectrum $E_n = -m_e e^4/(2(4\pi\varepsilon_0)^2\hbar^2 n^2)$ as derived input, compute the Balmer wavelengths and compare them with the observed hydrogen lines. Apply the electric-dipole rule to identify forbidden transitions.

Exercise 5.8 (Molecular symmetry)**.** For the equilibrium geometry of H_2O, determine the point-group action on nuclear displacements, remove translations and rotations, and decompose the vibrational representation into irreducibles. Determine infrared and Raman activity from the transformation types of the dipole and polarizability. Compare the predicted number and symmetry of lines with the observed three fundamental vibrational bands.

Part III

Fields

6 Manifolds, Forms, and Metrics

Definition 6.1 (Topological and smooth manifolds)**.** A *topological manifold* of dimension n is a Hausdorff, second-countable space locally homeomorphic to $\mathbb{R}^n$. A *smooth manifold* is a topological manifold with a maximal atlas whose transition maps are smooth. A map of smooth manifolds is *smooth* when its chart representatives are smooth.

Definition 6.2 (Tangent and cotangent spaces)**.** The *tangent space* T_pM is the vector space of derivations $v : C^\infty(M) \to \mathbb{R}$ at p, satisfying $v(fg) = f(p)v(g) + g(p)v(f)$. The *cotangent space* is $T_p^*M = (T_pM)^*$. The differential of $F : M \to N$ is $dF_p(v)(f) = v(f \circ F)$.

Exercise 6.1 (Intrinsic tangent vectors)**.** Prove that derivations at p, equivalence classes of smooth curves through p, and the usual coordinate tangent vectors are naturally isomorphic. Prove the chain rule $d(G \circ F)_p = dG_{F(p)} \circ dF_p$ and compute tangent spaces of regular level sets.

Definition 6.3 (Vector fields and flows)**.** A *vector field* is a smooth section $X : M \to TM$. A *flow* of X is a smooth local action Φ of $\mathbb{R}$ on M satisfying $\frac{d}{dt}\Big|_0 \Phi_t(p) = X_p$. The *Lie bracket* is the vector field $[X, Y](f) = X(Yf) - Y(Xf)$.

Exercise 6.2 (Flows and brackets)**.** Prove local existence and uniqueness of flows. Prove that $[X, Y] = 0$ if and only if their local flows commute, and that the Lie derivative $\mathcal{L}_X Y = [X, Y]$ is natural under diffeomorphisms.

Definition 6.4 (Exterior algebra and differential forms)**.** The *exterior algebra* of a vector space V is $\Lambda V = T(V)/\langle v \otimes v : v \in V\rangle$. A *differential k-form* on M is a smooth section of $\Lambda^k T^*M$. The *pullback* of $\omega \in \Omega^k(N)$ by $F : M \to N$ is

$$(F^*\omega)_p(v_1, \ldots, v_k) = \omega_{F(p)}(dF_p v_1, \ldots, dF_p v_k).$$

Definition 6.5 (Exterior derivative)**.** The *exterior derivative* is the unique family of maps $d : \Omega^k(M) \to \Omega^{k+1}(M)$ satisfying $df(X) = Xf$, the graded Leibniz rule, $d^2 = 0$, and naturality $F^*d = dF^*$.

Exercise 6.3 (Cartan calculus)**.** Construct d locally and prove its uniqueness. For contraction ι_X, prove Cartan's identity $\mathcal{L}_X = d\iota_X + \iota_X d$ and the naturality of d, wedge products, and pullbacks.

Definition 6.6 (Integration of forms)**.** An *orientation* of an n-manifold is a continuous choice of component of nonzero elements in $\Lambda^n T_p^* M$. The integral $\int_M \omega$ of a compactly supported top form is defined using orientation-preserving charts and a partition of unity.

Exercise 6.4 (Stokes' theorem)**.** Prove that the integral is independent of charts and partition of unity. For an oriented manifold with boundary, prove

$$\int_M d\omega = \int_{\partial M} \omega.$$

Deduce the fundamental theorem of calculus, Green's theorem, the divergence theorem, and the classical curl theorem.

Definition 6.7 (Tensor fields and metrics)**.** A (r, s)*-tensor field* is a smooth section of $TM^{\otimes r} \otimes T^*M^{\otimes s}$. A *Riemannian metric* is a smooth positive-definite inner product on each T_pM. A *Lorentzian metric* is a smooth nondegenerate symmetric form of signature $(-, +, \ldots, +)$. The length of a curve in a Riemannian manifold is $\int \sqrt{g(\dot\gamma, \dot\gamma)}\, dt$; the proper time of a timelike curve is $c^{-1} \int \sqrt{-g(\dot\gamma, \dot\gamma)}\, dt$. A metric and an orientation define a volume form vol_g.

Exercise 6.5 (Metric volume and proper time)**.** Construct vol_g intrinsically and derive its coordinate expression. Prove invariance of length and proper time under reparametrization. In Minkowski spacetime, maximize proper time among inertial paths with fixed endpoints and derive special-relativistic time dilation.

7 Connections, Curvature, and Field Equations

Definition 7.1 (Connections and geodesics)**.** A *connection* on TM is a map $\nabla : \Gamma(TM) \times \Gamma(TM) \to \Gamma(TM)$, linear over smooth functions in the first variable and satisfying $\nabla_X(fY) = X(f)Y + f\nabla_X Y$. Its torsion is $T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y]$. A *geodesic* satisfies $\nabla_{\dot\gamma}\dot\gamma = 0$. A connection is *metric compatible* if $Xg(Y,Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$.

Exercise 7.1 (Levi–Civita connection)**.** Prove that every Riemannian or Lorentzian metric has a unique torsion-free, metric-compatible connection. Derive the Koszul formula and prove that its geodesics are precisely the critical curves of the energy functional with fixed endpoints.

Definition 7.2 (Covariant derivative and parallel transport)**.** For a curve γ, the connection induces a *covariant derivative* D/dt on vector fields along γ. A field V is *parallel* when $DV/dt = 0$; the resulting linear isomorphism between endpoint tangent spaces is *parallel transport*.

Exercise 7.2 (Parallel transport and holonomy)**.** Prove existence, uniqueness, composition, and inverse properties of parallel transport. Show that Levi–Civita transport preserves the metric. Compute the rotation obtained by transporting a tangent vector around a latitude on the unit sphere and relate it to the enclosed curvature.

Definition 7.3 (Curvature)**.** The *curvature* of a connection is

$$R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.$$

For a metric connection, the *Riemann tensor* is $\mathrm{Riem}(X,Y,Z,W) = g(R(X,Y)Z,W)$. Its contraction is the *Ricci tensor* Ric; the contraction of Ricci is the *scalar curvature* S.

Exercise 7.3 (Curvature identities and geodesic deviation)**.** Prove tensoriality and the algebraic symmetries of the Riemann tensor and the first and second Bianchi identities. For a variation through geodesics with separation field J, prove

$$\frac{D^2 J}{dt^2} + R(J,\dot\gamma)\dot\gamma = 0.$$

Deduce the leading relative acceleration of nearby freely falling bodies.

Definition 7.4 (Smooth bundles and sections)**.** A *smooth fiber bundle* with fiber F is a smooth map $\pi : E \to M$ locally isomorphic over M to $U \times F \to U$. A *smooth vector bundle* has vector-space fibers and linear transition maps. A *smooth principal G-bundle* has a free right G-action, transitive on each fiber, and local equivariant trivializations. A *section* is a smooth map $s : M \to E$ with $\pi s = \mathrm{id}_M$. For a representation $G \to GL(V)$, the *associated vector bundle* is $P \times_G V = (P \times V)/((pg,v) \sim (p,gv))$.

Definition 7.5 (Gauge fields)**.** Let G be a Lie group and $\pi : P \to M$ a principal G-bundle. A *gauge connection* is an equivariant $\mathfrak{g}$-valued one-form A on P reproducing fundamental vertical fields. Its *curvature* is $F_A = dA + \frac{1}{2}[A \wedge A]$. A *gauge transformation* is a G-equivariant bundle automorphism covering id_M.

Exercise 7.4 (Gauge covariance and Bianchi identity)**.** Choose local sections s_i and derive transition maps $g_{ij} : U_i \cap U_j \to G$ and the transformation laws for the local potentials $A_i = s_i^* A$ and curvatures $F_i = s_i^* F_A$. Prove $d_A F_A = 0$ and gauge invariance of holonomy up to conjugacy. For $G = U(1)$, recover $A \mapsto A + d\chi$ and $F = dA$.

Exercise 7.5 (Yang–Mills equations and instantons). For $G = SU(r)$, use $\langle X, Y\rangle = -2\,\mathrm{Tr}(XY)$ and define

$$S_{\mathrm{YM}}(A) = \frac{1}{2g^2}\int_M \langle F_A \wedge *F_A\rangle$$

on a closed oriented Riemannian four-manifold. Prove gauge invariance and derive $d_A^* F_A = 0$, where $d_A^* = - * d_A *$. Decompose $F_A = F_A^+ + F_A^-$ and, for

$$k(A) = -\frac{1}{8\pi^2}\int_M \mathrm{Tr}(F_A \wedge F_A),$$

prove $S_{\mathrm{YM}}(A) \geq 8\pi^2|k(A)|/g^2$, with equality precisely for $F_A = \pm * F_A$. Use the Bianchi identity to show that these instantons solve the Yang–Mills equation.

Definition 7.6 (Electromagnetic field). On an oriented Lorentzian four-manifold, an *electromagnetic field strength* is a two-form F. Given a current one-form J and constants μ_0, c, *Maxwell's equations* are

$$dF = 0, \qquad d{*}F = \mu_0 {*}J.$$

Locally, a *gauge potential* is a one-form A with $F = dA$.

Exercise 7.6 (Maxwell theory and electromagnetic waves). Derive charge conservation from Maxwell's equations. On Minkowski spacetime, recover the electric and magnetic vector equations and the Lorentz force from F. In vacuum, impose Lorenz gauge and derive wave equations with speed c. Prove transverse polarization and compare the predicted relation $c = (\mu_0\varepsilon_0)^{-1/2}$ with the measured speed of light.

Definition 7.7 (Einstein equation). The *Einstein tensor* of a Lorentzian metric is $G = \mathrm{Ric} - \frac{1}{2}Sg$. A *stress–energy tensor* is a symmetric two-tensor T whose contraction with an observer's velocity gives the observer's energy–momentum current. The *Einstein equation* with cosmological constant Λ is

$$G + \Lambda g = \frac{8\pi G_N}{c^4} T.$$

Exercise 7.7 (Conservation and the Newtonian limit). Use the contracted Bianchi identity to derive $\nabla^a T_{ab} = 0$. Linearize $g = \eta + h$ for a static weak field and recover Poisson's equation and the Newtonian inverse-square law with the stated coupling constant.

Exercise 7.8 (Gravitational radiation). Linearize the vacuum Einstein equation about Minkowski spacetime, impose transverse–traceless gauge, and obtain two tensor polarizations satisfying a wave equation. Apply geodesic deviation to a ring of freely falling test masses and derive the strain measured by an orthogonal-arm interferometer.

Exercise 7.9 (Gravitational-wave detection). For a quasi-circular binary, derive the leading quadrupole waveform and the chirp relation between frequency, frequency derivative, and chirp mass. Using public values for GW150914, infer the chirp mass and order of magnitude of the strain. Compare these predictions with the measured coincident LIGO strain, limiting the conclusion to agreement of the waveform with general relativity in the observed regime.

Part IV

States

8 Hamiltonian States

Definition 8.1 (Configuration and phase spaces)**.** A *configuration space* is a smooth manifold Q. Its *phase space* is the cotangent bundle T^*Q. The tautological one-form on T^*Q is $\theta_\alpha(v) = \alpha(d\pi(v))$, and its canonical symplectic form is $\omega = -d\theta$.

Definition 8.2 (Symplectic dynamics)**.** A *symplectic form* is a closed, nondegenerate two-form ω. The *Hamiltonian vector field* of $H \in C^\infty(M)$ is defined by $\iota_{X_H}\omega = dH$. Its flow is the *Hamiltonian flow*. The *Poisson bracket* is $\{f, g\} = \omega(X_f, X_g)$.

Exercise 8.1 (Hamilton's equations and Poisson algebra)**.** Prove that the canonical form on T^*Q is symplectic. In canonical coordinates, derive Hamilton's equations. Prove antisymmetry, the Leibniz rule, and the Jacobi identity for the Poisson bracket, and prove $\dot{f} = \{H, f\} + \partial_t f$ along Hamiltonian flow.

Definition 8.3 (Symplectic actions and momentum maps)**.** An action of a Lie group G on (M, ω) is *symplectic* if it preserves ω. A *momentum map* is an equivariant map $J : M \to \mathfrak{g}^*$ such that $d\langle J, \xi\rangle = \iota_{\xi_M}\omega$ for every $\xi \in \mathfrak{g}$.

Exercise 8.2 (Noether's theorem)**.** Prove that if a Hamiltonian is invariant under a symplectic G-action with momentum map J, then every component of J is conserved. Compute the momentum maps for translations and rotations of a particle and recover linear and angular momentum conservation.

Exercise 8.3 (Liouville's theorem)**.** On a $2n$-dimensional symplectic manifold, prove that Hamiltonian flow preserves ω, the phase-space volume $\omega^n/n!$, and every density depending only on H. Deduce that Hamiltonian evolution cannot compress phase volume.

9 Probability, Gibbs States, and Ergodicity

Definition 9.1 (Probability)**.** A *probability space* is a measure space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{P}(\Omega) = 1$. A *random variable* is a measurable map. Its *expectation* is its integral. The *conditional expectation* $\mathbb{E}[X \mid \mathcal{G}]$ is the $\mathcal{G}$-measurable integrable random variable satisfying $\int_G \mathbb{E}[X \mid \mathcal{G}]\, d\mathbb{P} = \int_G X\, d\mathbb{P}$ for all $G \in \mathcal{G}$. Families of σ-algebras are *independent* when the probability of every finite intersection is the product of its probabilities.

Exercise 9.1 (Radon–Nikodym)**.** For finite measures $\nu \ll \mu$, prove that $\nu(A) = \int_A f\, d\mu$ for a unique $f \in L^1(\mu)$, $f \geq 0$. Extend the statement to finite signed measures.

Exercise 9.2 (Expectation, conditioning, and independence)**.** Use Radon–Nikodym to prove existence and almost-everywhere uniqueness of conditional expectation. Prove the tower property and conditional Jensen inequality. Characterize independence by factorization of expectations and derive variance addition for independent square-integrable random variables.

Definition 9.2 (Entropy and Gibbs ensembles)**.** For a discrete probability vector p, its *entropy* is $S(p) = -k_B \sum_i p_i \log p_i$. For a phase-space probability density ρ relative to Liouville measure μ_L, its *statistical entropy* is $S(\rho) = -k_B \int \rho \log \rho \, d\mu_L$. The *canonical Gibbs ensemble* at inverse temperature β is $d\mathbb{P}_\beta = Z(\beta)^{-1} e^{-\beta H} d\mu_L$.

Definition 9.3 (Measure-preserving dynamics and ergodicity)**.** A measurable transformation T is *measure preserving* if $\mu(T^{-1}A) = \mu(A)$. It is *ergodic* if every invariant measurable set has measure zero or full measure. An *equilibrium state* is an invariant probability measure for the dynamics.

Exercise 9.3 (Entropy and equilibrium)**.** Maximize entropy under normalization and fixed mean energy to derive the Gibbs ensemble. Prove $\langle H \rangle = -\partial_\beta \log Z$ and $\mathrm{Var}(H) = \partial_\beta^2 \log Z$. Prove invariance of Gibbs measure under Hamiltonian flow and formulate the consequence of the ergodic theorem for equality of time and ensemble averages.

Exercise 9.4 (Ideal-gas thermodynamics)**.** Compute the classical canonical partition function of N indistinguishable noninteracting particles in a box, including the phase-space normalization $h^{3N} N!$. Derive the ideal-gas law, internal energy, and entropy, and compare the pressure law with dilute-gas measurements within the classical regime.

10 Quantum and Algebraic States

Definition 10.1 (Quantum states and observables)**.** A bounded operator T is *trace class* when $\sum_n \langle e_n, |T| e_n \rangle < \infty$ for an orthonormal basis, where $|T| = (T^*T)^{1/2}$; this condition is basis-independent. For positive T, the sum defines $\operatorname{Tr} T$. A *density operator* is positive and trace class with $\operatorname{Tr} \rho = 1$. An *observable* is a self-adjoint operator A with spectral measure E_A. The *Born probability* of an event B is $\operatorname{Tr}(\rho E_A(B))$. A closed-system *unitary dynamics* is $\rho \mapsto U_t \rho U_t^*$ for a strongly continuous unitary group U_t. For unbounded observables and generators, spectral measures, expectations, and exponentials use the earlier unbounded spectral calculus and its natural domains.

Exercise 10.1 (Canonical quantization and Weyl relations)**.** On $\mathcal{S}(\mathbb{R}^n) \subset L^2(\mathbb{R}^n)$, set $Q_j \psi = x_j \psi$ and $P_j \psi = -i\hbar \partial_j \psi$. Prove $[Q_j, P_k] = i\hbar \delta_{jk}$ and derive the Weyl relations for their unitary groups. Using symmetric ordering, verify the Poisson-bracket–commutator rule for affine and quadratic observables. Compare different orderings of $q^2 p^2$ and explain the Groenewold–van Hove obstruction to quantizing all observables by this rule.

Exercise 10.2 (Quantized harmonic oscillator)**.** For $H = P^2/(2m) + m\omega^2 Q^2/2$, define creation and annihilation operators and prove $[a, a^*] = 1$ and $H = \hbar\omega(a^*a + 1/2)$. Construct the eigenstates with $E_n = \hbar\omega(n + 1/2)$ and use the spectral-lines exercise to obtain the frequency spacing ω.

Exercise 10.3 (Born rule and unitary dynamics)**.** Prove that the Born rule defines a probability measure and that its expected value is $\operatorname{Tr}(\rho A)$ when defined. Derive the von Neumann equation from $U_t = e^{-itH/\hbar}$. Prove conservation of expected energy and derive the Heisenberg equation for observables.

Exercise 10.4 (Stern–Gerlach projections and the Born rule)**.** For a spin-1/2 system and unit vector n, diagonalize $S_n = (\hbar/2) n \cdot \sigma$ and obtain $P_\pm(n) = (1 \pm n \cdot \sigma)/2$. For $\rho = (1 + r \cdot \sigma)/2$, derive $p_\pm(n) = \operatorname{Tr}(\rho P_\pm(n)) = (1 \pm r \cdot n)/2$. Model an inhomogeneous magnetic field by $H_{\mathrm{int}} = -\boldsymbol{\mu} \cdot \mathbf{B}$ and explain the separation into discrete beams. For successive analyzers along axes n and m, derive the transition probability $\operatorname{Tr}(P_+(n) P_+(m)) = (1 + n \cdot m)/2$ and interpret the intermediate projection. Compare the two-way splitting with Gerlach–Stern, *Zeitschrift für Physik* **9** (1922). Distinguish the original observation of space quantization in silver atoms from its later spin interpretation, and state which discrete outcomes and frequency predictions test the projection postulate and Born rule.

Exercise 10.5 (Blackbody radiation from quantized Fourier modes)**.** For electromagnetic radiation in a large conducting cavity, use its Fourier modes and two polarizations to derive the mode density $g(\omega) = \omega^2/(\pi^2 c^3)$ per unit volume. Show that classical Gibbs equipartition assigns mean energy $k_B T$ to each mode and produces the ultraviolet divergence. Using the oscillator levels derived in Exercise 10.2, compute the quantum Gibbs partition function and the thermal mean energy $\hbar\omega/(e^{\beta\hbar\omega} - 1)$. Deduce Planck's law

$$u(\omega, T) = \frac{\hbar\omega^3}{\pi^2 c^3} \frac{1}{e^{\hbar\omega/(k_B T)} - 1},$$

the Stefan–Boltzmann T^4 law, and Wien scaling $\omega_{\max} \propto T$. Compare the low- and high-frequency limits with the cavity spectra discussed by Planck, *Annalen der Physik* **309** (1901), and identify precisely where quantization removes the classical divergence.

Definition 10.2 (Composite and reduced states)**.** The state space of a composite system is $H_A \otimes H_B$. A state is *separable* if it lies in the closed convex hull of product density operators and *entangled* otherwise. The *reduced state* $\rho_A = \operatorname{Tr}_B \rho$ is determined by $\operatorname{Tr}(\rho_A A) = \operatorname{Tr}(\rho(A \otimes \mathrm{id}))$ for all bounded A.

Exercise 10.6 (Schmidt decomposition and entanglement)**.** Prove the Schmidt decomposition for pure vectors in a tensor product of finite-dimensional Hilbert spaces. Prove that a pure state is separable exactly when its reduced state has rank one, and that the two reduced states have the same nonzero spectrum. Compute the reduced states and entanglement entropy of a Bell state.

Definition 10.3 (C^*-algebras and states)**.** A *Banach algebra* is a complete normed algebra with $\|ab\| \le \|a\|\|b\|$. An *involution* satisfies $(ab)^* = b^* a^*$ and $(a^*)^* = a$. A *C^*-algebra* is an involutive Banach algebra satisfying $\|a^* a\| = \|a\|^2$. An element is *positive* if it is $b^* b$ for some b. A *state* is a positive linear functional φ with $\|\varphi\| = \varphi(1) = 1$.

Exercise 10.7 (Commutative Gelfand duality)**.** For compact Hausdorff X, prove that $C(X)$ is a commutative unital C^*-algebra. Prove that every commutative unital C^*-algebra A is isometrically $*$-isomorphic to $C(\operatorname{Spec} A)$ by the Gelfand transform, and identify states on $C(X)$ with probability measures.

Definition 10.4 (Representations and GNS construction)**.** A *representation* of a C^*-algebra A is a $*$-homomorphism $\pi : A \to B(H)$. For a state φ, define $N_\varphi = \{a : \varphi(a^* a) = 0\}$ and $\langle [a], [b] \rangle = \varphi(a^* b)$ on A/N_φ.

Exercise 10.8 (GNS theorem)**.** Complete A/N_φ and prove that left multiplication induces a cyclic representation $(\pi_\varphi, H_\varphi, \Omega_\varphi)$ satisfying $\varphi(a) = \langle \Omega_\varphi, \pi_\varphi(a)\Omega_\varphi \rangle$. Prove uniqueness up to a cyclic unitary equivalence.

Definition 10.5 (CAR algebra and fermionic Fock space)**.** For a complex Hilbert space K, the *canonical anticommutation relations algebra* $\mathrm{CAR}(K)$ is the unital C^*-algebra generated by conjugate-linear symbols $a(f)$ satisfying

$$\{a(f), a(g)\} = 0, \qquad \{a(f), a(g)^*\} = \langle f, g \rangle 1.$$

The *antisymmetric Fock space* is $\mathcal{F}_-(K) = \bigoplus_{n \ge 0} \Lambda^n K$. Its vacuum is $\Omega = 1 \in \Lambda^0 K$; creation by f is exterior multiplication, and annihilation by f is its adjoint.

Exercise 10.9 (CAR and GNS predict fermionic antibunching)**.** Prove that creation and annihilation on $\mathcal{F}_-(K)$ satisfy the CAR and that $\omega_0(x) = \langle \Omega, x\Omega \rangle$ is a state. Apply the preceding exercise to identify its GNS representation with the Fock representation. Let ω_C be a gauge-invariant quasifree state with one-particle density $0 \le C \le 1$, and use its determinantal Wick rule to prove, for orthonormal detector modes f, g, that

$$\omega_C(N_f N_g) = \omega_C(N_f)\omega_C(N_g) - |\langle f, Cg \rangle|^2, \qquad N_f = a(f)^* a(f).$$

Deduce $N_f^2 = N_f$ and fermionic antibunching. Compare the sign and separation dependence of the correlation deficit with the neutral-atom measurements in Fölling et al., *Nature* **444** (2006).

Definition 10.6 (Operator topologies and von Neumann algebras)**.** The *weak operator topology* is generated by $T \mapsto \langle x, Ty \rangle$; the *strong operator topology* is generated by $T \mapsto \|Tx\|$. For $S \subseteq B(H)$, its *commutant* is $S' = \{T : TS = ST \text{ for all } S \in S\}$. A *von Neumann algebra* is a unital $*$-subalgebra $M \subseteq B(H)$ with $M = M''$.

Exercise 10.10 (Bicommutant theorem)**.** Prove that for a unital $*$-subalgebra $A \subseteq B(H)$, the strong and weak operator closures of A equal A''. Deduce that von Neumann algebras have sufficient projections to approximate their self-adjoint elements spectrally.

Definition 10.7 (Normality and spatial tensor products)**.** A positive functional φ on a von Neumann algebra is *normal* if $a_\lambda \uparrow a$ implies $\varphi(a_\lambda) \uparrow \varphi(a)$ for every bounded increasing net of positive elements. A representation is *normal* if it preserves such suprema. For concrete von Neumann algebras $M \subseteq B(H)$ and $N \subseteq B(K)$, their *spatial tensor product* is

$$M \mathbin{\overline{\otimes}} N = (M \odot N)'' \subseteq B(H \otimes K).$$

Exercise 10.11 (Normal states and independent subsystems)**.** Show that every density operator ρ gives the normal state $a \mapsto \mathrm{Tr}(\rho a)$, and, using trace-class duality, prove the converse on $B(H)$. Prove $B(H) \mathbin{\overline{\otimes}} B(K) = B(H \otimes K)$ and identify product states and partial traces in this language. Explain why local normality—normality on every local algebra—is the infinite-system analogue of requiring ordinary reduced density matrices for all finite laboratories.

Definition 10.8 (Cyclic and separating vectors; modular Hamiltonians)**.** For $M \subseteq B(H)$, a vector Ω is *cyclic* if $\overline{M\Omega} = H$ and *separating* if $a\Omega = 0$ implies $a = 0$. For a faithful density operator ρ, its *modular Hamiltonian* is $K_\rho = -\log \rho$ and its modular flow on $B(H)$ is $\sigma_t(a) = \rho^{it} a \rho^{-it}$.

Exercise 10.12 (Finite modular theory and entanglement tomography)**.** For finite-dimensional H, let $B(H)$ act on itself by left multiplication with inner product $\langle X, Y \rangle = \mathrm{Tr}(X^* Y)$, and set $\Omega_\rho = \rho^{1/2}$ for faithful ρ. Prove that Ω_ρ is cyclic and separating. For $S_0(a\Omega_\rho) = a^*\Omega_\rho$, compute the polar decomposition and show that its modular operator is $\Delta(X) = \rho X \rho^{-1}$, yielding the flow above. If ρ is a reduced state, recover its entanglement spectrum from K_ρ. Compare this reconstruction with the trapped-ion data analyzed by Kokail et al., *Nature Physics* **17** (2021), and explain why finite-subsystem tomography does not by itself establish continuum modular theory.

Definition 10.9 (Traces, factors, and noncommutative probability)**.** A *trace* on a von Neumann algebra is a normal positive functional τ satisfying $\tau(ab) = \tau(ba)$. A *factor* is a von Neumann algebra with center $\mathbb{C}1$; it is *type I* if it is isomorphic to some $B(K)$. A *noncommutative probability space* is a pair (M, φ) of a von Neumann algebra and a normal state; its random variables are operators affiliated with M.

Exercise 10.13 (Projection dimension and factors)**.** For a finite factor with normalized trace τ, prove additivity and unitary invariance of τ on projections. Relate the possible projection values to matrix factors and to diffuse finite factors, and formulate independence of commuting subalgebras by factorization of the state.

11 Completely Bounded Structure and Correlations

Definition 11.1 (Positive and completely positive maps)**.** A linear map $\Phi : A \to B$ between C^*-algebras is *positive* if it preserves positive elements and *completely positive* if every $\mathrm{id}_{M_n} \otimes \Phi$ is positive. A *quantum channel* is a completely positive trace-preserving map on trace-class operators, equivalently a unital completely positive map on observables in the adjoint picture.

Exercise 11.1 (Stinespring and Kraus representations)**.** Prove that a unital completely positive map $\Phi : A \to B(H)$ has a representation $\Phi(a) = V^*\pi(a)V$. In finite dimensions, derive a Kraus representation $\Phi(a) = \sum_i K_i^* a K_i$ and its normalization. Prove that every channel is unitary evolution on a larger system followed by discarding an environment.

Exercise 11.2 (Amplitude damping reproduces spontaneous emission)**.** For a qubit with ground state $|0\rangle$, excited state $|1\rangle$, and environment vacuum $|0\rangle_E$, let

$$\begin{aligned} V_t|0\rangle &= |0\rangle|0\rangle_E, \\ V_t|1\rangle &= \sqrt{\eta_t}\,|1\rangle|0\rangle_E + \sqrt{1-\eta_t}\,|0\rangle|1\rangle_E, \qquad \eta_t = e^{-t/T_1}. \end{aligned}$$

Trace out the environment and derive

$$K_0 = |0\rangle\langle 0| + \sqrt{\eta_t}|1\rangle\langle 1|, \qquad K_1 = \sqrt{1-\eta_t}|0\rangle\langle 1|.$$

Prove complete positivity, trace preservation, and the semigroup law. Derive $\rho_{11}(t) = e^{-t/T_1}\rho_{11}(0)$ and $\rho_{01}(t) = e^{-t/(2T_1)}\rho_{01}(0)$. Compare the predicted decay and the environmental dependence of T_1 with the transmon measurements in Houck et al., *Physical Review Letters* **101** (2008).

Definition 11.2 (Measurements)**.** A *positive operator-valued measure* is a countably additive map E from measurable events to positive operators with $E(X) = 1$. In state ρ, its outcome probability is $\mathrm{Tr}(\rho E(B))$.

Exercise 11.3 (POVMs and dilation)**.** Prove that projection-valued measures are POVMs. Prove Naimark's dilation theorem in finite dimensions and derive the measurement channel associated to a POVM. Show that state discrimination can require a nonprojective POVM.

Definition 11.3 (Operator spaces)**.** An *operator space* is a closed subspace $E \subseteq B(H)$ with matrix norms inherited from $M_n(E) \subseteq B(H^{\oplus n})$. A map $u : E \to F$ is *completely bounded* if $\|u\|_{cb} = \sup_n \|\mathrm{id}_{M_n} \otimes u\| < \infty$. The matrix order of an operator system is the family of positive cones $M_n(E)_+$.

Definition 11.4 (Ancilla-assisted channel discrimination)**.** A *one-use discrimination strategy* for channels $\Phi_1, \ldots, \Phi_m$ prepares a state on $H \otimes K$, applies $\Phi_j \otimes \mathrm{id}_K$ to the unknown channel input, and measures the output with a POVM. The strategy is *ancilla assisted* when $\dim K > 1$. It is *unambiguous* when every conclusive outcome identifies the channel without error.

Exercise 11.4 (An ancilla improves measurement-channel discrimination)**.** For $0 < \theta < \pi/4$, define

$$|\phi\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle, \quad |\psi\rangle = \cos\theta|0\rangle - \sin\theta|1\rangle,$$

and let $\mathcal{M}, \mathcal{N}$ be the two projective measurement channels with bases $\{\phi, \phi^\perp\}$ and $\{\psi, \psi^\perp\}$. Use a singlet probe, retain the second qubit as an ancilla, and condition on the classical measurement outcome. Reduce the task to unambiguous discrimination of $|\phi\rangle$ and $|\psi\rangle$ and prove

$$P_I^{\mathrm{anc}} = \cos(2\theta), \qquad P_I^{\mathrm{sep}} = \frac{1+\cos^2(2\theta)}{2}$$

for the optimal ancilla-assisted and single-qubit strategies. Deduce the strict advantage for $0 < \theta < \pi/4$ and express it as a distinction between the first and amplified matrix levels of the measurement channels. Compare the predicted success and inconclusive probabilities with Miková et al., *Physical Review A* **90** (2014).

Exercise 11.5 (Ruan's theorem)**.** Prove that concrete operator-space norms satisfy

$$\|x \oplus y\| = \max(\|x\|, \|y\|), \qquad \|axb\| \leq \|a\|\|x\|\|b\|.$$

Prove conversely that every vector space with matrix norms satisfying these axioms admits a complete isometry into $B(H)$.

Exercise 11.6 (Arveson–Wittstock extension)**.** Prove in finite dimensions that a completely positive map from an operator system extends completely positively to the containing C^*-algebra. Deduce, by the 2×2 operator-system construction, that completely bounded maps admit norm-preserving extensions.

Definition 11.5 (Operator-space tensor products)**.** The *minimal operator-space tensor norm* on $E \otimes F$ is inherited from concrete inclusions in $B(H \otimes K)$. The *Haagerup norm* is

$$\|z\|_h = \inf\{\|[x_1 \ \cdots \ x_r]\| \, \|[y_1 \ \cdots \ y_r]^T\| : z = \sum_i x_i \otimes y_i\}.$$

A bilinear form $u : E \times F \to \mathbb{C}$ is *jointly completely bounded* when its associated matrix amplifications are uniformly bounded.

Exercise 11.7 (Tensor norms linearize bilinear maps)**.** Prove the universal property of the projective Banach-space tensor norm. Prove that the Haagerup tensor product linearizes completely bounded factorizations and compare the minimal, Haagerup, and projective norms on finite-dimensional matrix spaces.

Definition 11.6 (Bell functionals and nonlocal games)**.** A *Bell functional* is an array $M = (M_{xy}^{ab})$ evaluated on conditional distributions by $\langle M, p\rangle = \sum_{a,b,x,y} M_{xy}^{ab} p(a, b \mid x, y)$. A two-player *nonlocal game* $G = (\pi, V)$ consists of a question distribution $\pi(x, y)$ and a payoff $0 \leq V(a, b \mid x, y) \leq 1$. Its classical and entangled values $\omega(G)$ and $\omega^*(G)$ are the optimal expected payoffs using, respectively, shared randomness and local POVMs $\{E_a^x\}_a, \{F_b^y\}_b$ on a shared finite-dimensional state ρ, with $p(a, b \mid x, y) = \mathrm{Tr}(\rho(E_a^x \otimes F_b^y))$. An *XOR game* has binary answers and scores only their parity, equivalently a correlation $\sum_{x,y} M_{xy}\mathbb{E}[a_x b_y]$ for signs a_x, b_y.

Exercise 11.8 (Magic-square pseudotelepathy)**.** The referee chooses $(x, y) \in \{1, 2, 3\}^2$ uniformly. Alice returns a row $a \in \{\pm 1\}^3$ with product $+1$, Bob returns a column $b \in \{\pm 1\}^3$ with product -1, and they win when $a_y = b_x$. Prove by multiplying all row and column constraints that no deterministic classical strategy wins every question, construct one losing on only one question, and hence show $\omega(G_{\mathrm{MS}}) = 8/9$. Using two Bell pairs and a Mermin–Peres square of commuting two-qubit Pauli observables, construct a strategy with $\omega^*(G_{\mathrm{MS}}) = 1$.

Compare the ideal query-by-query predictions with the hyperentangled-photon experiment of Xu et al., *Physical Review Letters* **129** (2022). Explain separately the locality and fair-sampling assumptions in that experiment. Show why a score above 8/9 is a device-independent witness under the stated assumptions; then, assuming a robust magic-square rigidity theorem, explain how a score $1 - \varepsilon$ certifies two Bell pairs up to local isometries and errors controlled by ε.

Definition 11.7 (The tensor of a game)**.** Set

$$E_{X,A} = \ell_1^X(\ell_\infty^A), \qquad \|z\|_{E_{X,A}} = \sum_x \max_a |z_{x,a}|,$$

with its standard dual operator-space structure. The tensor associated to $G = (\pi, V)$ is

$$\widehat{G} = \sum_{x,y,a,b} \pi(x,y)V(a,b \mid x,y)(e_x \otimes e_a) \otimes (e_y \otimes e_b) \in E_{X,A} \otimes E_{Y,B}.$$

Exercise 11.9 (Game values as Banach and operator-space norms)**.** Identify deterministic strategies with extreme points of the relevant unit balls and prove

$$\omega(G) = \|\widehat{G}\|_{E_{X,A}\otimes_\varepsilon E_{Y,B}}, \qquad \omega^*(G) = \|\widehat{G}\|_{E_{X,A}\otimes_{\min} E_{Y,B}}.$$

Equivalently, show that these are the bounded and completely bounded norms of the bilinear payoff form on the dual mixed-norm spaces. Locate in the proof the use of positivity of a game tensor, and explain why the same simple formula need not hold for an arbitrary signed Bell functional. Use Palazuelos–Vidick, *Survey on Nonlocal Games and Operator Space Theory*, *Journal of Mathematical Physics* **57** (2016) as a guide.

Exercise 11.10 (Bounded and unbounded quantum advantages)**.** For an XOR game, prove that its classical and entangled biases are

$$\beta(G) = \sup_{a_x, b_y = \pm 1} \left|\sum M_{xy} a_x b_y\right|, \quad \beta^*(G) = \sup_{\|u_x\|, \|v_y\| \le 1} \left|\sum M_{xy} \langle u_x, v_y\rangle\right|.$$

Use Tsirelson's vector representation and the real Grothendieck inequality to prove $\beta^*(G) \le K_G^{\mathbb{R}} \beta(G)$. For CHSH, compute the classical bound 2 and quantum bound $2\sqrt{2}$.

For general multi-output games, take as supplied the Khot–Vishnoi estimates $\omega(G_{\mathrm{KV}}) \le C/n$ and $\omega^*(G_{\mathrm{KV}}) \ge c/(\log n)^2$, achieved with local entanglement dimension n. Use the preceding norm dictionary to deduce

$$\frac{\|\widehat{G}_{\mathrm{KV}}\|_{\min}}{\|\widehat{G}_{\mathrm{KV}}\|_\varepsilon} \ge c' \frac{n}{(\log n)^2}.$$

Explain why the XOR comparison remains dimension independent while general game tensors require operator-space matrix levels. Consult Buhrman–Regev–Scarpa–de Wolf, *Theory of Computing* **8** (2012) for the supplied estimates.

Exercise 11.11 (Measured nonclassical correlations)**.** For a polarization-entangled two-photon state, choose four polarization angles attaining the CHSH value $2\sqrt{2}$ and derive the predicted coincidence correlations. Analyze a loophole-free Bell-test data table with its stated uncertainties and determine whether the classical bound is excluded, restricting the conclusion to local hidden-variable models satisfying the experimental assumptions.

Part V

Topology

12 Homotopy, Cohomology, and Bundles

Definition 12.1 (Homotopy and fundamental group)**.** A *homotopy* from $f : X \to Y$ to $g : X \to Y$ is a continuous map $H : X \times [0,1] \to Y$ with endpoints f and g. The *fundamental group* $\pi_1(X, x_0)$

is the group of endpoint-fixed homotopy classes of loops based at x_0, with concatenation.

Exercise 12.1 (Fundamental groups and winding)**.** Prove homotopy invariance of π_1 and the Seifert–van Kampen theorem for a union with path-connected intersection. Compute $\pi_1(S^1) \cong \mathbb{Z}$ by lifting loops to $\mathbb{R}$, and prove that the resulting integer is the winding number.

Definition 12.2 (Homology and cohomology)**.** The *singular chain group* $C_n(X; R)$ is the free R-module on continuous maps $\Delta^n \to X$, with alternating face boundary ∂. The *homology* is $H_n(X; R) = \ker \partial_n / \operatorname{im} \partial_{n+1}$. The *cochain complex* is $C^n(X; R) = \mathrm{Hom}(C_n(X; R), R)$ with coboundary $\delta\phi = \phi\partial$, and its cohomology is $H^n(X; R)$.

Definition 12.3 (Products and pairings)**.** The *Kronecker pairing* is $H^n(X; R) \times H_n(X; R) \to R$, $([\phi], [c]) \mapsto \phi(c)$. The *cup product* is the product on cohomology induced by the front–back face formula on cochains.

Exercise 12.2 (Homology, cohomology, and degree)**.** Prove $\partial^2 = 0$, $\delta^2 = 0$, and homotopy invariance. Compute the homology and cohomology rings of spheres and tori. For a continuous map $f : S^n \to S^n$, define its degree by $f_*[S^n] = \deg(f)[S^n]$ and prove multiplicativity and agreement with winding number for $n = 1$.

For the rest of this section, forget the smooth structure on the bundles introduced with gauge fields and regard their local trivializations and transition maps as continuous. We study them up to topological isomorphism.

Exercise 12.3 (Topological bundles from transition functions)**.** Prove that continuous transition functions satisfying the cocycle condition construct a topological bundle and that changes of local trivialization give isomorphic bundles. Prove that a principal bundle is trivial exactly when it has a global section. Construct associated vector bundles from representations of G.

Definition 12.4 (Global connections and holonomy)**.** A *connection* on a vector bundle $E \to M$ is a map $\nabla : \Gamma(E) \to \Omega^1(M; E)$ satisfying $\nabla(fs) = df \otimes s + f\nabla s$. Its curvature is $F_\nabla = \nabla^2 \in \Omega^2(M; \mathrm{End}\, E)$. Its *holonomy* along a loop is the automorphism of the initial fiber given by parallel transport.

Exercise 12.4 (Curvature and holonomy)**.** Derive local connection and curvature forms and their gauge-transformation laws. Prove the Bianchi identity. Show that infinitesimal holonomy is curvature and compute the $U(1)$ holonomy around the boundary of a surface as $\exp(i \int F)$ when a global potential is available.

Exercise 12.5 (Aharonov–Bohm holonomy)**.** On $X = \mathbb{R}^2 \setminus \{0\}$, consider the $U(1)$ gauge potential $A = (\Phi/2\pi)d\theta$. Verify that $F = dA = 0$ on X but $\oint_\gamma A = \Phi$ for a loop of winding number one. Show that a particle of charge q acquires the gauge-invariant phase $\exp(iq\Phi/\hbar)$, derive the relative phase of two paths enclosing the excluded flux, and obtain the flux period $h/|q|$. Prove that A cannot be removed by a single-valued gauge transformation unless its holonomy is trivial. Compare the predicted fringe displacement with the shielded-flux electron-holography experiment of Tonomura et al., *Physical Review Letters* **56** (1986). Explain why the result establishes the physical significance of holonomy, not of a gauge-dependent potential by itself, even though curvature vanishes along the electron paths.

Supplied input (Classifying spaces)**.** Let $BU(n) = \mathrm{Gr}_n(\mathbb{C}^\infty)$ be the infinite Grassmannian with tautological bundle $\gamma_n \to BU(n)$. We take as supplied the classification theorem that, for paracompact X, pullback gives a bijection between $[X, BU(n)]$ and isomorphism classes of rank-n complex vector bundles. We also take as supplied the universal classes $c_j(\gamma_n) \in H^{2j}(BU(n); \mathbb{Z})$.

Definition 12.5 (Chern classes and character)**.** If a rank-n complex vector bundle $E \to X$ is classified by $f : X \to BU(n)$, its *Chern classes* are $c_j(E) = f^* c_j(\gamma_n) \in H^{2j}(X; \mathbb{Z})$. They are natural and satisfy $c(E \oplus F) = c(E)c(F)$. The *Chern character* is the natural ring map $\mathrm{ch} : K^0(X) \to H^{\mathrm{even}}(X; \mathbb{Q})$ whose value on a line bundle L is $e^{c_1(L)}$.

Exercise 12.6 (Curvature representatives)**.** For a Hermitian bundle with unitary connection, prove that

$$\det\left(1 + \frac{i}{2\pi} F_\nabla\right) \quad \text{and} \quad \operatorname{Tr}\exp\left(\frac{i}{2\pi} F_\nabla\right)$$

are closed forms whose cohomology classes are independent of the connection. Identify them with the Chern classes and Chern character and prove their Whitney-sum formulas.

Definition 12.6 (Berry geometry)**.** For a smooth family of one-dimensional spectral projections P_x on a Hilbert space, the ranges form a line bundle. The *Berry connection* is $\nabla = P \circ d$ and its curvature is the *Berry curvature*. Its holonomy is the *Berry phase*. The integral $\langle c_1(L), [\Sigma] \rangle$ is the *Chern number* over an oriented closed surface Σ.

Exercise 12.7 (Berry phase and quantized response)**.** Compute the Berry curvature of a two-level Hamiltonian $H(n) = n \cdot \sigma$ over S^2 and its Chern number. For a gapped two-dimensional periodic Hamiltonian, derive the Kubo formula expressing Hall conductance as e^2/h times the occupied-band Chern number. Deduce integer quantization under gap-preserving perturbations and compare with integer quantum Hall plateaus.

13 K-Theory and Index

Definition 13.1 (Grothendieck completion and stable equivalence)**.** The *Grothendieck group* of a commutative monoid M is the universal abelian group receiving a monoid map from M. Vector bundles E, F are *stably equivalent* if $E \oplus \underline{\mathbb{C}}^m \cong F \oplus \underline{\mathbb{C}}^n$ for some m, n. The group $K^0(X)$ is the Grothendieck group of complex vector bundles on compact X; define $K^1(X) = \widetilde{K}^0(\Sigma X)$.

Exercise 13.1 (Bott periodicity and topological phases)**.** Prove the clutching description of bundles on spheres. Establish Bott periodicity in the form $\widetilde{K}^0(S^{n+2}) \cong \widetilde{K}^0(S^n)$ and compute the complex K-groups of spheres and tori. Show that a gapped family of finite-rank Hamiltonians defines the stable class of its occupied bundle and identify the class-A invariants in dimensions one through three.

Definition 13.2 (Operator K-theory)**.** For a unital C^*-algebra A, $K_0(A)$ is the Grothendieck group of stable homotopy classes of projections in matrix algebras over A, and $K_1(A)$ is the stable homotopy group of unitaries in matrix algebras over A.

Exercise 13.2 (Topological and operator K-theory)**.** Construct the Serre–Swan correspondence between vector bundles over compact X and finitely generated projective $C(X)$-modules. Deduce $K^j(X) \cong K_j(C(X))$ and formulate Bott periodicity for C^*-algebras.

Definition 13.3 (Compact and Fredholm operators)**.** An operator on a Hilbert space is *compact* if it is a norm limit of finite-rank operators. A bounded operator $T : H_0 \to H_1$ is *Fredholm* if its kernel and cokernel are finite-dimensional and its range is closed; its *index* is $\operatorname{ind} T = \dim\ker T - \dim\operatorname{coker} T$. The *Calkin algebra* is $\mathcal{Q}(H) = B(H)/\mathcal{K}(H)$.

Exercise 13.3 (Fredholm index and the Calkin algebra)**.** Prove that T is Fredholm exactly when its image in the Calkin algebra is invertible. Prove homotopy invariance, compact-perturbation invariance, and additivity of the index. Compute the index of the unilateral shift and identify the boundary map $K_1(\mathcal{Q}(H)) \to K_0(\mathcal{K}(H))$.

Supplied input (Index-theorem foundations)**.** We take as supplied the construction of Sobolev spaces and completions, elliptic estimates and elliptic regularity, and the extension of elliptic operators to Fredholm maps between Sobolev spaces. We also take as supplied spin structures (lifts of oriented orthonormal frame bundles from $SO(n)$ to $\mathrm{Spin}(n)$), their spinor bundles and Dirac operators, the characteristic class $\widehat{A}(TM) = \prod_j (x_j/2)/\sinh(x_j/2)$ in formal curvature roots, and the Thom isomorphism and K-theoretic pushforward. The following exercises apply this analytic and spin-geometric machinery rather than construct it.

Definition 13.4 (Elliptic symbols)**.** For vector bundles E, F over a compact smooth manifold, the *principal symbol* of an order-m differential operator $D : \Gamma(E) \to \Gamma(F)$ is the homogeneous bundle map $\sigma_D : \pi^*E \to \pi^*F$ on T^*M. The operator is *elliptic* if $\sigma_D(x, \xi)$ is invertible for every $\xi \neq 0$. Its *analytic index* is its Fredholm index on Sobolev completions; its *topological index* is the K-theoretic pushforward of its symbol class.

Exercise 13.4 (Atiyah–Singer index theorem)**.** Use the supplied elliptic estimates to deduce Fredholmness on a compact manifold. Establish the Atiyah–Singer equality $\mathrm{ind}_{an} D = \mathrm{ind}_{top}(\sigma_D)$, using Bott periodicity and the axiomatic characterization of the index. Derive the Riemann–Roch and Dirac-index formulas as special cases.

Definition 13.5 (Chirality and instanton number)**.** Let M be an oriented even-dimensional Riemannian spin manifold. The *chirality operator* splits its complex spinor bundle as $S = S^+ \oplus S^-$. For a Hermitian vector bundle E with unitary connection A, the coupled Dirac operator is odd and has a *chiral part*

$$D_A^+ : \Gamma(S^+ \otimes E) \longrightarrow \Gamma(S^- \otimes E).$$

For an instanton or anti-instanton from Exercise 7.5, its *instanton number* is

$$k = -\frac{1}{8\pi^2} \int_M \mathrm{Tr}(F_A \wedge F_A) \in \mathbb{Z}.$$

Exercise 13.5 (Fermion chirality detects instanton number)**.** Identify the characteristic number $k(A)$ from Exercise 7.5 with $\int_M \mathrm{ch}_2(E)$. Then apply the Dirac-index formula from the preceding exercise to prove

$$\mathrm{ind}\, D_A^+ = \dim\ker D_A^+ - \dim\ker(D_A^+)^* = \int_M \big[\widehat{A}(TM)\,\mathrm{ch}(E)\big]_4.$$

For the fundamental $SU(2)$ bundle over S^4, show that this index is k with the orientation convention in the definition. Use the Weitzenböck formula for a self-dual or anti-self-dual connection to determine which chirality has no zero modes and deduce that the other has exactly $|k|$ zero modes. For a fermion in a representation R, derive the factor $2T(R)k$. Finally, compute the fermionic measure under the axial rotation $\psi \mapsto e^{i\alpha\gamma_5}\psi$ and obtain the integrated anomaly $\Delta Q_5 = 2\,\mathrm{ind}\, D_A^+$, identifying the chirality change carried by an instanton transition.

Exercise 13.6 (Index theory and invertible free-fermion phases)**.** Let $H(k)$ be a continuous family of finite-dimensional self-adjoint free-fermion Hamiltonians over a compact momentum space X, with a uniform gap at zero. Flatten its spectrum to

$$Q(k) = \mathrm{sgn}\, H(k) = H(k)|H(k)|^{-1}, \qquad P(k) = \frac{1 - Q(k)}{2},$$

and prove that the occupied spaces $E_k = \operatorname{im} P(k)$ form a vector bundle $E \to X$. Show that its stable class $[E] \in K^0(X)$ is invariant under gap-preserving deformations.

When X is an even-dimensional spin manifold, pair this class with its Dirac operator and use Atiyah–Singer to obtain

$$\langle [E], [D_X] \rangle = \operatorname{ind}(D_X^+ \otimes E) = \int_X \widehat{A}(TX)\, \operatorname{ch}(E).$$

For a two-dimensional Brillouin torus, identify this integer with the first Chern number and derive the class-A Hall response $\sigma_H = (e^2/h)\langle c_1(E), [X] \rangle$.

Prove that stacking Hamiltonians gives direct sum and addition in $K^0(X)$. After passing to reduced K-theory, show that spectral reversal replaces E by $E^\perp$ and gives the additive inverse because $E \oplus E^\perp$ is trivial. Explain why an invertible phase can have protected boundary modes or symmetry defects but no nontrivial intrinsic bulk fusion sectors. Contrast this group-valued classification with the sector-valued noninvertible phases in the next section. With additional symmetries, formulate the replacement by the appropriate real, equivariant, or twisted K-group, following Freed–Moore, *Annales Henri Poincaré* **14** (2013).

14 Subfactors, Conformal Nets, and Tensor Categories

Definition 14.1 (Subfactors and conditional expectations)**.** A *subfactor* is an inclusion $N \subseteq M$ of factors with common unit. A *conditional expectation* $E : M \to N$ is a positive unital N-bimodule projection. For finite factors with normalized trace, the *Jones index* $[M : N]$ is the Murray–von Neumann dimension of $L^2(M)$ as a left N-module.

Definition 14.2 (Basic construction)**.** On $L^2(M)$, the *Jones projection* e_N is the orthogonal projection onto $L^2(N)$. The *basic construction* is $M_1 = \langle M, e_N \rangle''$. Iteration gives the *Jones tower* $N \subseteq M \subseteq M_1 \subseteq M_2 \subseteq \cdots$.

Exercise 14.1 (Jones tower and Temperley–Lieb relations)**.** For finite-index $N \subseteq M$, derive $e_N x e_N = E(x) e_N$ and construct the canonical trace on the basic construction. Prove that the normalized Jones projections in the tower satisfy

$$e_i^2 = e_i, \qquad e_i e_j = e_j e_i \ (|i-j| > 1), \qquad e_i e_{i\pm1} e_i = [M : N]^{-1} e_i.$$

Exercise 14.2 (Diagrammatic calculus and index quantization)**.** Construct the Temperley–Lieb diagrammatic category, with composition by stacking and each closed loop valued at $\delta = [M : N]^{1/2}$. Use positivity of its Markov trace to prove that index values below 4 are $4\cos^2(\pi/n)$ for integers $n \geq 3$.

Definition 14.3 (Braid groups and Markov traces)**.** The *braid group* B_n has generators σ_i with relations $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$ and $\sigma_i \sigma_j = \sigma_j \sigma_i$ for $|i-j| > 1$. A *Markov trace* on a tower of braid-group quotients is a compatible trace satisfying a fixed stabilization rule.

Exercise 14.3 (Jones polynomial)**.** Construct a braid-group representation from the Temperley–Lieb generators. Use the Markov trace and Markov moves to obtain an invariant of oriented links, normalize it to the Jones polynomial, and derive its skein relation. Compute it for the unknot, trefoil, and Hopf link.

Definition 14.4 (Local conformal nets)**.** Let $\mathcal{I}$ be the proper open intervals of S^1. A *local conformal net* assigns a von Neumann algebra $\mathcal{A}(I) \subset B(H)$ to each $I \in \mathcal{I}$ such that

$$I \subset J \Longrightarrow \mathcal{A}(I) \subset \mathcal{A}(J), \qquad I \cap J = \varnothing \Longrightarrow [\mathcal{A}(I), \mathcal{A}(J)] = 0.$$

The assignment is covariant under orientation-preserving diffeomorphisms of S^1, represented projectively on H with positive rotation generator. It has an invariant vacuum vector Ω, cyclic for the local algebras, and $\bigvee_I \mathcal{A}(I) = B(H)$. Physically, S^1 is the compactified chiral light ray and $\mathcal{A}(I)$ consists of observables localized in I. The norm closure of $\bigcup_I \mathcal{A}(I)$ is the *quasilocal algebra.*

Definition 14.5 (Duality and regularity for conformal nets)**.** Write I' for the interior of the complement of I. A net has *Haag duality* when $\mathcal{A}(I) = \mathcal{A}(I')'$. It has the *split property* when $\overline{I} \subset J$ implies $\mathcal{A}(I) \subset F \subset \mathcal{A}(J)$ for some type-I factor F. It is *strongly additive* when removing an interior point from I to obtain $I_1 \cup I_2$ gives $\mathcal{A}(I) = \mathcal{A}(I_1) \vee \mathcal{A}(I_2)$. A representation of the quasilocal algebra is *locally normal* when its restriction to every $\mathcal{A}(I)$ is normal.

Exercise 14.4 (The two-interval subfactor)**.** For $E = I_1 \cup I_2$ a union of disjoint intervals, set $\mathcal{A}(E) = \mathcal{A}(I_1) \vee \mathcal{A}(I_2)$. First use locality and vacuum cyclicity for $\mathcal{A}(I')$ to prove that the vacuum is separating for $\mathcal{A}(I)$. Use the split property and the spatial tensor product to identify separated local algebras as independent subsystems. Then use locality to obtain the inclusion

$$\mathcal{A}(E) \subseteq \mathcal{A}(E')'.$$

Under the split and irreducibility hypotheses, show that this is a subfactor. Define the *μ-index* $\mu(\mathcal{A}) = [\mathcal{A}(E')' : \mathcal{A}(E)]$, and prove that it is independent of the two-interval configuration by conformal covariance. Contrast single-interval Haag duality $\mathcal{A}(I) = \mathcal{A}(I')'$ with the possible failure of duality for E, measured by this Jones index.

Definition 14.6 (DHR sectors)**.** A *DHR sector* of $\mathcal{A}$ is represented by a locally normal, transportable endomorphism ρ of the quasilocal algebra that acts identically on $\mathcal{A}(I')$ for some interval I; transportability means that an equivalent representative can be localized in any interval. Composition is the tensor product; transporting two localized endomorphisms past one another gives their braiding. For a finite-index irreducible sector localized in I, its statistical dimension satisfies

$$d(\rho)^2 = [\mathcal{A}(I) : \rho(\mathcal{A}(I))].$$

Supplied input (DHR and complete-rationality theorems)**.** We take as supplied that charge transporters for localized finite-index DHR endomorphisms produce a unitary braiding. We also use the theorem that a split, strongly additive conformal net with finite μ-index has finitely many irreducible sectors, nondegenerate braiding, and $\mu(\mathcal{A}) = \sum_i d(\rho_i)^2$; see Kawahigashi–Longo–Müger, *Communications in Mathematical Physics* **219** (2001).

Exercise 14.5 (Critical Ising chain, Rydberg spectroscopy, and the Ising net)**.** For $J, h > 0$, consider the transverse-field Ising chain

$$H_{\mathrm{I}} = -J \sum_i Z_i Z_{i+1} - h \sum_i X_i.$$

Use the Jordan–Wigner transformation and a Bogoliubov rotation to obtain

$$\varepsilon(k) = 2\sqrt{J^2 + h^2 - 2Jh\cos k}.$$

Show that the gap is $2|J-h|$ and closes at $h=J$, where the long-wavelength modes are massless Majorana fermions. Taking the standard fermionic determinant asymptotics as input, obtain $\langle Z_i Z_{i+r}\rangle \sim Cr^{-1/4}$ and a connected energy-density correlator proportional to r^{-2}. Identify the bulk Ising fields $1, \sigma, \varepsilon$ with scaling dimensions $0, 1/8, 1$, respectively.

The Rydberg experiment realizes a different microscopic chain,

$$H_{\mathrm{R}} = \sum_i \left(\frac{\Omega}{2} P_{i-1} X_i P_{i+1} - \Delta n_i\right) + \sum_{j-i\geq 2} V_{ij} n_i n_j,$$

in the nearest-neighbor blockade regime. Explain why equality of microscopic Hamiltonians is unnecessary for the two critical points to share the Ising universality class; here n_i projects onto the Rydberg state and $P_i = 1 - n_i$. Take as input the open-chain CFT spectrum, including its conformal towers and $E_a(L) - E_0(L) = \pi v x_a/L + o(L^{-1})$, and recover the universal even-parity ratios $2:4:6:8$ and odd-parity ratios $3:5:7$. Compare them with the reported modulation-spectroscopy peaks of Sun et al., *Experimental observation of conformal field theory spectra* (2026), stating the finite-size and experimental limitations.

Identify the chiral continuum theory with the $c = 1/2$ Ising conformal net. Its DHR sectors $1, \psi, \sigma$ have conformal weights $0, 1/2, 1/16$; relate the bulk energy field to $\psi\bar{\psi}$ and distinguish these chiral weights from the bulk scaling dimensions above. Explain why the experiment supports the Ising CFT spectrum but does not directly measure the net or all of its superselection sectors.

Definition 14.7 (Monoidal and rigid categories)**.** A *monoidal category* is a category $\mathcal{C}$ with a tensor functor, unit object, and coherent associativity and unit isomorphisms. A *dual* of X is an object X^* with evaluation and coevaluation morphisms satisfying the snake identities. A monoidal category is *rigid* if every object has left and right duals.

Definition 14.8 (Fusion, braided, and modular categories)**.** A *fusion category* over $\mathbb{C}$ is a finite semisimple rigid $\mathbb{C}$-linear monoidal category with simple unit. It is *unitary* when it is a C^*-category and its tensor product and structural isomorphisms are compatible with the adjoint. A *braiding* is a natural family $c_{X,Y} : X \otimes Y \to Y \otimes X$ satisfying the hexagon identities. A *unitary modular tensor category* is a unitary ribbon fusion category with unitary nondegenerate braiding.

Exercise 14.6 (From Jones index to a modular category)**.** Assume that $\mathcal{A}$ is completely rational: it is split and strongly additive, with finite μ-index. Apply the supplied complete-rationality theorem to obtain finitely many irreducible DHR sectors ρ_i with nondegenerate braiding and

$$\mu(\mathcal{A}) = \sum_i d(\rho_i)^2.$$

Conclude that the sectors form a unitary modular tensor category. For the Ising net at $c = 1/2$, take the sectors $1, \psi, \sigma$ with dimensions $1, 1, \sqrt{2}$ and fusion rules

$$\psi^2 = 1, \qquad \psi\sigma = \sigma, \qquad \sigma^2 = 1 \oplus \psi.$$

Compute $\mu(\mathcal{A})$ and interpret ψ and σ as the chiral Majorana and spin sectors.

Exercise 14.7 (Fusion rules and braiding)**.** Prove that simple-object classes of a fusion category form a based ring under tensor product. Derive the pentagon and hexagon equations for associativity and braiding data. Compute quantum dimensions and braid-group representations for the Fibonacci fusion rule $\tau \otimes \tau \cong 1 \oplus \tau$.

Definition 14.9 (Chern–Simons theory and Wilson loops)**.** For a connection A on a principal bundle over an oriented three-manifold and an invariant inner product on $\mathfrak{g}$, the *Chern–Simons functional* at level k is

$$S_{CS}(A) = \frac{k}{4\pi} \int_M \left\langle A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right\rangle.$$

For a representation V and closed curve K, the *Wilson loop* is $W_V(K) = \mathrm{Tr}_V(\mathrm{Hol}_A(K))$.

Exercise 14.8 (Gauge invariance, level, and knot invariants)**.** Compute the change of S_{CS} under a gauge transformation and derive the integrality condition on k required for $e^{iS_{CS}}$ to be gauge invariant. Derive flatness as the field equation. Relate Wilson-loop expectations for $SU(2)$ to the Jones polynomial at a root of unity.

Definition 14.10 (Bordisms and topological field theories)**.** The *bordism category* has closed oriented $(n-1)$-manifolds as objects, oriented n-dimensional bordisms as morphisms, and disjoint union as tensor product. An *n-dimensional topological field theory* is a symmetric monoidal functor from this category to vector spaces. An extended theory may assign categories to codimension-two manifolds. A TQFT is *invertible* when its state spaces are tensor-invertible (hence one-dimensional over a field) and every bordism acts invertibly; noninvertible TQFTs may instead carry nontrivial superselection and fusion sectors.

Exercise 14.9 (Functorial gluing)**.** Show that the monoidal-functor axioms give multiplicativity under disjoint union, composition under gluing, duality under orientation reversal, and mapping-class-group representations on state spaces. In two dimensions, prove the equivalence with commutative Frobenius algebras.

Definition 14.11 (Anyons and topological degeneracy)**.** In a two-dimensional gapped phase, an *anyon type* is a simple object of its braided fusion category. It is *abelian* if its quantum dimension is 1 and *nonabelian* otherwise. The *topological degeneracy* on a closed surface is the dimension of the state space assigned by the associated topological field theory.

Exercise 14.10 (Abelian and nonabelian anyons)**.** Compute fusion spaces and braid actions for an abelian category and for the Fibonacci category. Prove that nonabelian fusion spaces can have dimension greater than one and that braiding acts projectively on them. Compute the torus degeneracy from the simple anyon types in a modular category.

Exercise 14.11 (Quantum Hall phases and measured topological order)**.** Use flux insertion for a Laughlin state at filling $\nu = 1/m$ to derive quasiparticle charge e/m, exchange phase π/m, Hall conductance $\nu e^2/h$, and genus-g degeneracy m^g. Compare these consequences separately with conductance plateaus, fractional shot-noise charge measurements, and interferometric phase data, stating which topological predictions each observation tests and which it does not.